

Squeeze Every Bit of Insight: Leveraging Few-shot Models with a Compact Support Set for Domain Transfer in Object Detection from Pineapple Fields

1st Fabián Fallas-Moya
Atlantic Campus
Universidad de Costa Rica
Cartago, Costa Rica
fabian.fallas-moya@ucr.ac.cr

2nd Danny Xie-Li
Computer Science Department
Instituto Tecnológico de Costa Rica
Cartago, Costa Rica
dxie@ic-itcr.ac.cr

3rd Saúl Calderón-Ramírez
Computer Science Department
Instituto Tecnológico de Costa Rica
Cartago, Costa Rica
sacalderon@itcr.ac.cr

Abstract—Object Detection (OD) typically requires large labeled datasets and significant resources. To address this, we propose a novel two-stage pipeline that combines Visual Foundation Models (VFM) for object proposals with few-shot models, using Mahalanobis distance with support and context prototypes. This method improves upon Euclidean-based approaches by incorporating data covariance into the prototypes. Validated on pineapple detection using drone imagery, our approach not only outperforms state-of-the-art (SOTA) few-shot models with minimal labeled samples but also provides a practical solution for supporting farmers in monitoring and managing their crops. Extensive experiments show that FastSAM, paired with a Mahalanobis distance variant using SVD and diagonal loading for regularization, achieves the highest mAP for drone imagery, surpassing other methods.

Index Terms—few-shot, object detection, agriculture, mahalanobis

I. INTRODUCTION

In computer vision, the Object Detection (OD) task is of significant importance. Usually, it involves two primary phases: first, locating objects through boxes, and second, classifying these boxes into distinct categories [1]. The disadvantage of OD is that it requires a substantial number of labels for training. Some datasets typically require several hours to be fully labeled [2], [3]. This significant expense associated with labeling introduces several disadvantages to adopting Deep Learning (DL) in the industry [4]. This study proposes the use of a few-shot model along with a Visual Foundation Models (VFM) – which has been trained using millions of samples –. This can simplify the task of OD, rendering the training phase unnecessary while using very small labeled datasets to build the prototypes.

In the proposed method, we leverage the power of VFMs as the *proposal generator* for OD. Then, we used few-shot models to classify these proposals (see Fig. 1). This two-stage process is similar to Faster R-CNN [5] since we perform two stages; however, in our proposal, we rely on two components: (i) VFMs for the first stage, leveraging their power to segment the entire image, and (ii) few-shot models [6] using a small number of annotated images to classify the segments,

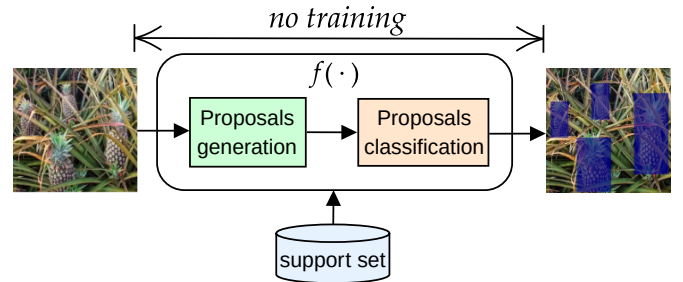


Fig. 1: We perform OD without the backpropagation of the regular training. This method uses two stages: the first stage generates object proposals using VFMs, and the second stage classifies these proposals using few-shot models (using only a small number of labels to create the support set).

using prototype-based few-shot learning. This approach makes object detection applications in fields such as agriculture more efficient, as considerably less training is needed. To demonstrate this, we tested the proposed method using drone images of pineapple crops from a large agricultural company (Upala Agrícola, located in Costa Rica).

Some approaches use a few-shot model that utilizes prior knowledge to differentiate observations when the labeled dataset is limited or small. For example, using the Euclidean distance with an Euclidean prototype [4] yielded promising results. However, a notable limitation of Euclidean distance is the uniform treatment of all dimensions and the assumption of independence. To address this limitation, we suggest refining the usage of the Mahalanobis distance for classifying object proposals to improve the performance of zero-shot OD.

The calculation of the Mahalanobis distance relies on estimating the multivariate normal density of feature embeddings generated by pre-trained models. However, when dealing with high-dimensional data as well as limited observations in few-shot settings, the accuracy of parametric estimates of multivariate normal density over the embeddings becomes unstable owing to the insufficient number of observations, which affects

the effectiveness of the Mahalanobis distance [7]. To address this, we propose the use of Singular Value Decomposition (SVD) to reduce the dimensionality of the embeddings using the Mahalanobis distance in our few-shot prototypes. In the results section, we show the impact of this procedure.

In general, DL relies on having large datasets, as a result, we propose a *no-training* approach that leverages VFMs and few-shot models to perform OD. Using domain adaptation [8]–[10], our method outperforms SOTA approaches that use prototypical networks [11] and density prototypes [4].

We summarize the contributions of this work as follows:

- We present a novel pipeline using VFMs and few-shot models with a *no training* policy, which improves implementation feasibility in precision agriculture applications.
- We use the Mahalanobis distance as the distance metric for the few-shot models. An additional innovation involves using the support set to create a primary prototype (pineapple) and extracting sections from the background to build the *context* prototype.
- We propose the usage of SVD to reduce the dimensionality of the embedding space.
- Finally, we show a statistical analysis to find the best configuration, analyzing different levels and factors.

II. RELATED WORK

Object Detection. Zou et al. [12] developed a review of OD and its application to different fields, including agriculture. Girshick [5] proposed Faster R-CNN, which significantly impacted OD performance using a two-stage model. Approaches using the YOLO family [13] (single-stage detectors) have been proposed. However, in OD, the labeling process is expensive [14] and alternatives have been explored in recent years, such as semi-supervised learning [15] or few-shot learning [16], using small labeled sets.

Object Detection Applications to Precision Agriculture. Zhang and Wu [2] show an early attempt to use Machine Learning (ML) to classify fruits using Support Vector Machines (SVM). Kamilaris and Prenafeta [17] demonstrate the improvement in the performance of agricultural applications using Convolutional Neural Networks (CNN). However, this method requires thousands of images for training. Klompenburg et al. [18] conducted an interesting survey of approximately 537 studies on crop yield prediction. However, most studies did not consider OD or segmentation problems. Regarding OD, DeepFruits [19] presented a simple framework that showed good results, and another study used feature pyramid networks [14], showing good application in OD; however, these works require a significant number of images during training.

Few-shot Models. The Siamese Neural Network [20] was the first successful attempt to use few-shot DL models with acceptable results. Subsequently, prototypical networks [11] showed good results when using the Euclidean distance to class prototypes. Density prototypes [4] are another interesting approach, because more statistical information is used in the classification. Similarly, Talur et al. [21] used Mahalanobis

[22] to perform few-shot using a small support set. Bateni et al. [23] proposed the use of Mahalanobis for image classification using a regularization term. However, none applied a few-shot to OD.

Visual Foundation Models. There has been a notable shift in the OD field owing to the emergence of the VFMs. These models, trained on large datasets with reduced supervision—utilizing techniques such as self- or semi-supervised learning, and even incorporating multimodal data—have significantly reshaped the computer vision landscape. One such model is the Segment Anything Model (SAM) ([24]). Leveraging extensively pre-trained models presents the opportunity to require less supervision data from the target domain [25], [26]. A compelling approach to harnessing these pre-trained models is the implementation of few-shot, as proposed by Le-Cacheux et al. [27].

III. PROPOSED METHOD

Traditional training requires data and resource-intensive backpropagation. This work proposes an alternative pipeline, $f(\cdot)$, without backpropagation. It simulates the classic two-stage OD, where it first finds proposals (using a VFM, and then classifies these proposals (using few-shot models). Before the execution of $f(\cdot)$, we first created the few-shot prototypes using the *prototype creation* $h(\cdot)$ method. The descriptions of $h(\cdot)$ and $f(\cdot)$ are as follows.

A. Prototype creation $h(\cdot)$

Fig. 2 (bottom) illustrates the prototype creation process, $h(\cdot)$. A few annotated images ($s_1, s_2, \dots, s_N$) are used to extract the bounding boxes from the ground truth images to create the support set $\hat{S}_s$. Some surrounding regions are cropped to create the context class $\hat{S}_c$, i.e., sections of the images that do not have the objects of interest. To estimate $\hat{S}_c$, region proposals are generated using selective search [29]. These proposals are restricted to having less than 30% overlap with the support class samples ($\hat{S}_s$). The context class $\hat{S}_c$ is important because it simulates false-positive objects that can potentially be detected by SAM, because SAM segments the entire image, including objects that are not the target. Therefore, having a context class is beneficial for detecting false-positive objects.

The cropped regions (represented as a matrix $\hat{S} \in \mathbb{R}^{N \times D}$, where N is the number of samples and D is the dimensionality of each sample) are processed through a feature extraction backbone, denoted by $g : \mathbb{R}^{N \times D} \rightarrow \mathbb{R}^{N \times M}$, where M is the dimensionality of the embedding space. This backbone maps each sample to a lower-dimensional embedding space of fixed size.

In order to create our prototypes, we used three key elements: SVD, covariance matrices, and Inter-Quartile Range (IQR) to create a parameter to be used during classification.

1) *Singular Value Decomposition:* When the support set has fewer examples than dimensions ($N < D$), estimating the sample covariance matrix becomes unstable [30] and may result in a singular matrix [31]. This instability can

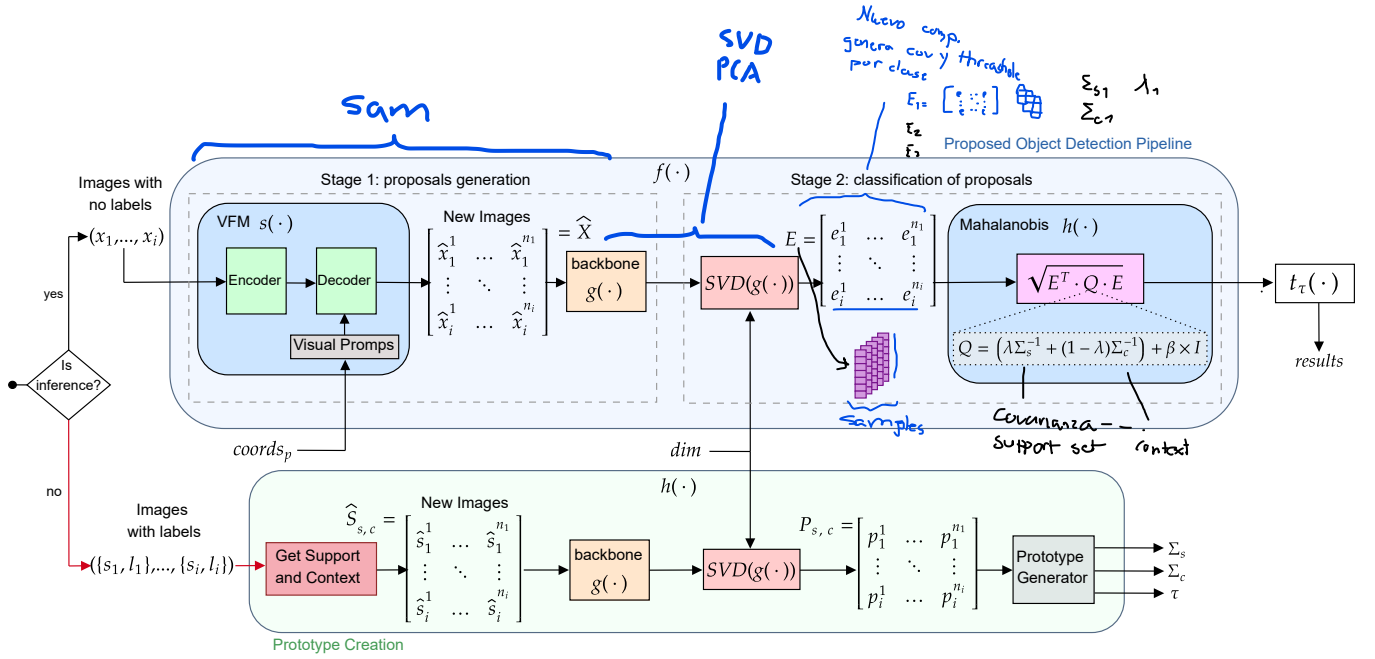


Fig. 2: In our proposal, we perform two main actions: prototype creation $h(\cdot)$ and object detection $f(\cdot)$. In the prototype creation $h(\cdot)$ we create the covariance matrices Σ_s and Σ_c that will work as our prototypes (also the parameter τ to classify samples) to perform few-shot classification. Then, function we perform OD $f(\cdot)$ following a two-stage process [28]: (1) a VFM generates object proposals, and (2) these proposals are classified into their respective classes using the prototypes.

lead to inconsistent outcomes, particularly in Mahalanobis distance calculations. To address this issue, we propose using SVD to reduce the dimensionality of the feature embeddings, ensuring $N > D$, thereby stabilizing the covariance estimates and improving the robustness of the Mahalanobis distance computation.

SVD is rarely used in deep learning due to one main reason: it has a high computational complexity of $O(d^3)$ for square matrices [32], and if backpropagation is used, the process can be complex [33]. Since we are not using backpropagation, we compute the SVD to reduce dimensionality, enhancing computational efficiency and enabling faster processing in this case. The function $SVD(\cdot)$ generates a new set of embeddings.

The lower-dimensional embeddings (denoted by p) are subsequently used to estimate the parameters of a multivariate Gaussian distribution by constructing covariance matrices Σ_s and Σ_c for the support and context prototypes, respectively. The support prototype, represented by Σ_s corresponds to the target class (i.e., pineapples). The context prototype, represented by Σ_c includes both the target and its surrounding regions, providing additional contextual information about the target. These prototypes are used in our classifier (Sec. III-B2).

2) *Covariance Matrices*: To estimate the class sample covariance matrices Σ_s (support) and Σ_c (context), we directly compute them using the dimensionality-reduced feature embeddings from $SVD(g(\cdot))$. The covariance matrix for each class $k \in \{s, c\}$ is calculated as follows:

$$\Sigma_k = \frac{1}{N-1} \sum_{i=1}^N (p_i - \mu_k)(p_i - \mu_k)^T \quad (1)$$

where $k \in \{s, c\}$ denotes the support (s) and context (c) classes. N represents the number of observations (or objects in our case), and p_i is the i -th dimensionality-reduced feature embedding. The mean vector of class k , denoted by μ_k , is computed as $\mu_k = \frac{1}{N} \sum_{i=1}^N p_i$ (as the regular mean over the samples).

3) *A parameter to filter results (λ)*: In addition to calculating the covariance matrices, we derive a threshold τ for the support set using the IQR method. We take all final embeddings p and we calculate their distance from the prototype Σ_s , so we can get distances from all support samples. With these distances, we can get the IQR and our τ . We use this method to classify an observation x into a class k .

The IQR is defined as: $IQR = Q_3 - Q_1$, where Q_1 represents the first quartile and Q_3 signifies the third quartile of the computed distances. We derive a threshold parameter τ as: $\tau = Q_3 + (1.5 \times IQR)$. To classify a new unseen sample x_i , we use the method t_τ as:

$$t_\tau(x_i) = \begin{cases} 0 & \text{if } d \geq \tau \\ 1 & \text{if } d < \tau \end{cases} \quad (2)$$

where d is the distance for a new sample x_i . If d exceeds the threshold τ , the sample is classified as 0 (not an object) or 1 (object found). This method is used at the end of the OD pipeline shown in Fig. 2.

B. Proposed Object Detection Pipeline $f(\cdot)$

Our OD pipeline, denoted by $f(\cdot)$ and illustrated in Fig. 2-top, consists of two main stages: proposal generation and classification of these proposals, which is similar to the Faster R-CNN framework [28]. Our pipeline integrates VFM, feature

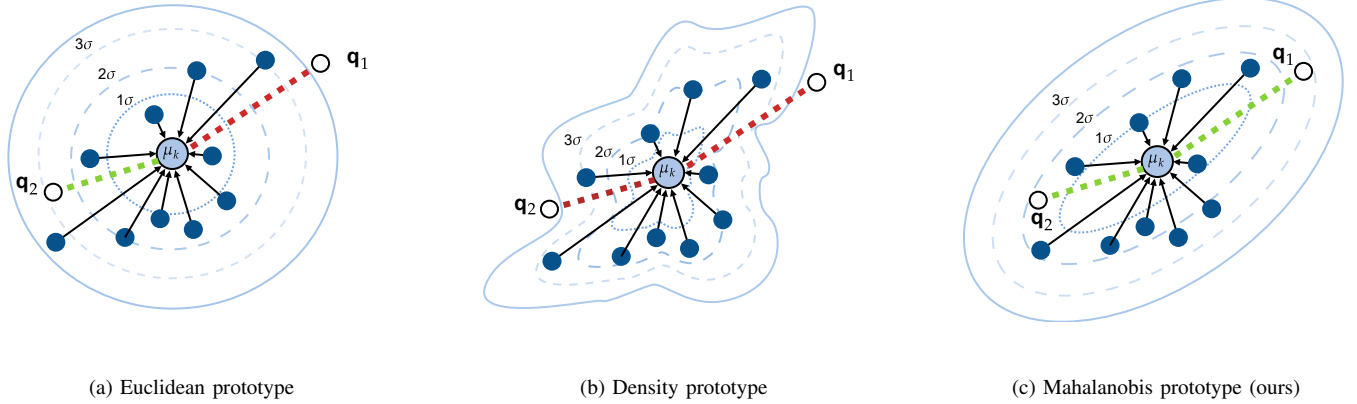


Fig. 3: Few-shot prototypes are generated using the support set (blue dots), and two queries q_1 and q_2 from the same class are evaluated. Red dotted lines indicate misclassifications, while green lines show correct classifications. Fig. **a** shows the Euclidean prototype method, where q_1 is misclassified due to the prototype’s lack of sensitivity to data distribution. Fig. **b** depicts a density-based approach with per-dimension density estimates; both queries are misclassified due to overfitting. Fig. **c** illustrates our Mahalanobis distance-based method, which uses the data covariance to form more representative prototypes, leading to correct classification of both queries.

extractors, few-shot methods, and SVD for dimensionality reduction. This combination provides a cost-effective solution that avoids the need for extensive training with backpropagation.

1) *Proposals Generation*: In the first stage of the OD pipeline (Fig. 2-top) processes an initial set of unlabeled images $(x_1, \dots, x_i)$ through a VFM module, denoted by $s(\cdot)$. This module leverages SAM variants, such as SlimSAM, MobileSAM, EdgeSAM, HQ-SAM, and FastSAM. These variants operate using user-provided visual prompts (e.g., text, coordinates, or bounding boxes). In our implementation, we used a grid of 1024 points as a visual prompt. In particular, FastSAM employs YOLOv8-seg for all-instance segmentation to generate mask coefficients. The resulting segmentation masks identify the regions of interest ($\hat{x}$), which are then cropped from the original image x (as bounding boxes).

Next, using the cropped regions ($\hat{x}$), we create a matrix $\hat{X}$ (Fig. 2-top, Stage 1) with the dimensions of $\mathbb{R}^{N \times D}$, where N is the number of samples and D is the dimensionality of each sample. This matrix is mapped into a M -dimensional feature space by using a feature extractor $g : \mathbb{R}^{N \times D} \rightarrow \mathbb{R}^{N \times M}$. The resulting feature embedding matrix is denoted by $\mathbb{M}^{N \times D}$, where D is the dimensionality of the embeddings of all N samples. These feature embeddings are then used in the next stage for proposal classification using few-shot methods.

2) *Classification of Proposals*: We apply SVD to the embeddings from the previous stage using the same dimension used when creating the prototypes Σ_s and Σ_c (Sec. III-A). These prototypes or covariance matrices are used to compute the distance between the sample x and the prototype. The classification is based on the distance to the prototype by using the parameter τ (Sec. III-A3). However, preliminary tests showed poor performance owing to the limited number

of samples available for constructing Σ_s and Σ_c , which often results in singular, non-invertible covariance matrices.

To address this, we incorporate diagonal loading [30] alongside SVD, improving the robustness of covariance estimation. This technique, similar to the regularization approach proposed by Bateni et al. [23], stabilizes the estimation process. To our knowledge, this is the first application of diagonal loading in this context. The final formula is given by:

$$Q = (\lambda \Sigma_s^{-1} + (1 - \lambda) \Sigma_c^{-1}) + \beta \times I, \quad (3)$$

where Σ_s is the estimated covariance matrix for the support class s , Σ_c is the estimated covariance matrix for the context class c , I is the identity matrix used for diagonal loading, and the parameters λ and β control the influence of the context samples and the degree of diagonal loading, respectively.

Finally, to classify new samples, we employ Q as one of the terms to calculate the Mahalanobis distance. Our formula incorporates a sample x (sample to be classified), a mean vector μ , and the covariance matrix Q (Fig. 3c). Our final formula is defined as:

$$D_k(x) = \sqrt{E^T \cdot Q \cdot E}, \quad (4)$$

where E represents $\text{SVD}(g(\hat{X}))$, the samples to be classified, and Q is the covariance-based metric from Eq. 3. Notably, Eq. 4 does not require μ , as our formulation leverages Q to encode class structure directly, improving robustness and eliminating the need for explicit mean subtraction (typically required in the Mahalanobis distance). Finally, we use Eq. 2 to classify samples using the threshold τ .

Fig. 3a illustrates the baseline approach using Euclidean space, which creates a uniform prototype according to the distances in the support set. Additionally, the density method

[4] in Fig. 3b uses the support set to generate density functions, for every dimension in the embedding space, under the Gaussian distribution assumption, though it risks overfitting as the data size decreases. Our proposed method, shown in Fig. 3c, leverages the Mahalanobis distance and data covariance. As demonstrated in our experiments, the Mahalanobis approach improves performance compared to the other two metrics, emphasizing the importance of fitting the data to account for its structure.

IV. EXPERIMENTS

We start our evaluation by finding the best number of SVD components. After identifying the optimal configuration, we fine-tune our method using the Mahalanobis distance. Finally, we compare the performance of our approach against SOTA methods to assess its effectiveness.

Dataset: The dataset consists of 155 manually labeled high-resolution images captured by drones flying at an altitude of approximately 15 meters over a commercial pineapple plantation in Costa Rica. These images were acquired from Upala Agrícola, the world’s largest producer of fresh pineapples [34], as part of a tropical agriculture monitoring initiative. The aerial imagery enables detailed analysis of crop distribution and plant health at scale, making it particularly valuable for precision agriculture applications. By leveraging drone-based data collection, this dataset supports the development of automated object detection systems that can enhance crop monitoring, optimize resource allocation, and assist farmers in making data-driven decisions for more efficient and sustainable agricultural practices.

From these images, 100 were designated for testing, and 55 were available to build the support set. Images were acquired using a multispectral DJI Phantom 4 drone with RTK technology.

A. Dimensionality Reduction

First, we explored how varying the number of SVD components affects the Mahalanobis prototype (see Fig. 4). Using the XCiT model with a 128-dimensional vector embedding and FastSAM for object proposals, we compared scenarios with and without SVD for dimensionality reduction. We tested with a reduction size of 128, 64, 32, 16, 8, and 4 components. We use 1–5 labeled drone images (every image has approximately 20 pineapples), and evaluated performance using Mean Average Precision (mAP) with a 0.5 Intersection over the Union (IoU) threshold.

As shown in Fig. 4, convergence is observed with 16 and 32 components as the number of labeled images increases, and with 64 components when using five labeled images. Notably, reducing the number of components enhances performance, particularly in low-data scenarios. With only one labeled image, using 4 components yields an average mAP of 0.387 ($\sigma = 0.01$), whereas 8 components results in a similar mAP of 0.383 ($\sigma = 0.01$). In contrast, without dimensionality reduction, the performance deteriorates significantly, achieving only 0.277 mAP ($\sigma = 0.114$) even with five labeled images.

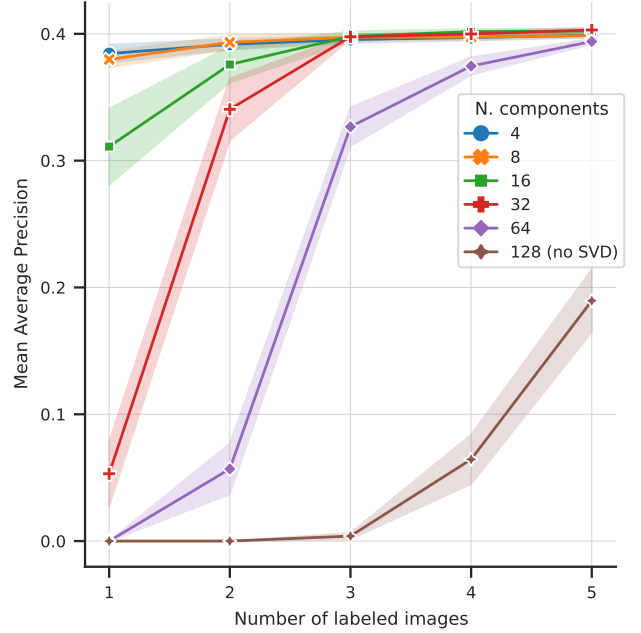


Fig. 4: Interaction with standard error (shaded region), showing the relationship between the number of labeled images, the number of components, and the Mean Average Precision (mAP).

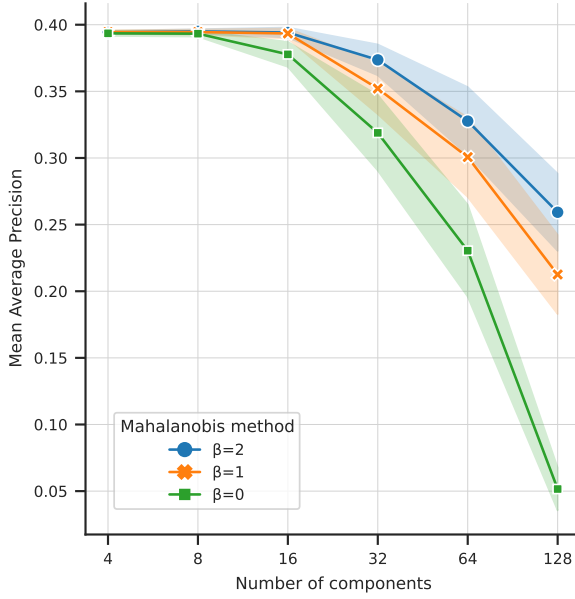
These results highlight the advantages of lower-dimensional representations for improving model robustness and consistency.

As the dimensionality of the feature embedding increases with a few samples, the covariance matrix becomes increasingly sparse. This sparsity poses a challenge in capturing correlations among dimensions, resulting in instability when the Mahalanobis distance method is used. Inversely, as the dimensionality decreases, the data becomes less sparse through dimensionality reduction, leading to more stable results in calculating the distance metric. Therefore, we demonstrate the significance of SVD in enhancing the efficacy of Mahalanobis distance.

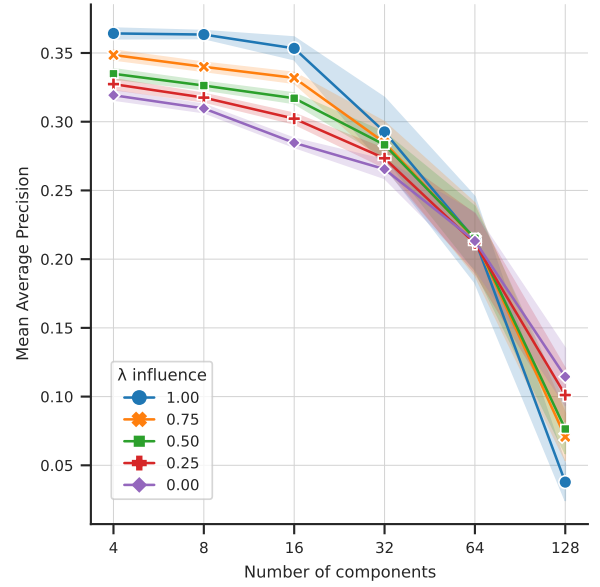
B. Fine Tuning λ and β

A thorough analysis of the influence of λ and β – from Equation 3 – was conducted, taking into account the number of k components used for embedding dimensionality reduction. The objective is to identify the optimal values for λ , β , and the number of k components (SVD) that maximize the mAP metric at a 0.5 IoU threshold.

1) Analyzing the Influence of β : We examined the effect of diagonal loading in Equation 3, where I is the identity matrix and β is the diagonal loading value. As Carlson [30] noticed, small eigenvalues can cause numerical instability, which diagonal loading helps address by adjusting eigenvalues. We tested $\beta = 0, 1, 2$ (no loading, moderate, and high loading) across component sizes 4, 8, 16, 32, 64, 128 using the XCiT model as our feature extractor and Fast SAM as the proposal



(a) β influence and the number of components.



(b) λ influence and the number of components.

Fig. 5: Interaction plots of different factors showing the standard errors (shaded region) to analyze the influence of β and λ in Eq. 3 and to obtain the best combination to create out final Mahalanobis prototype.

model, with 5 replicates per setup. Results in Fig. 5a demonstrate that reducing the number of components diminishes the effect of β , resulting in behavior similar to cases without β influence. This phenomenon is notably evident when utilizing 4 or 8 components.

This finding indicates that with a small number of samples, the covariance matrix can become ill-conditioned, with eigenvalues near zero or negative, causing instability. Introducing positive β values to the diagonal improves conditioning and stability. However, this effect lessens as the number of components decreases and labeled images increase. Thus, balancing diagonal loading and component count is crucial for effective results.

2) **Analyzing the influence of λ :** To analyze λ , we evaluate its impact across different component counts. Our methodology uses SAM as the proposal model and the XCiT model as the feature extractor, again.

Fig. 5b shows the impact of $\lambda = 1$ across different component counts. As components increase, the influence of $\lambda = 1$ grows, surpassing other λ values. We also note that $\lambda = 0$ is beneficial with 128 components, but this advantage decreases with fewer components.

Table I highlights the significance of Eq. 3 by illustrating the effect of λ on the number of objects. As expected, performance improves as this number increases. With fewer objects, the context matrix ($\lambda \approx 0$) plays a more prominent role; however, the high variance suggests no statistically significant difference. Notably, with a limited number of objects (around 50), the available data is insufficient to achieve acceptable results, emphasizing the need for more samples. In contrast, as the number of objects grows, the influence of $\lambda = 1$

steadily increases, outperforming other values of λ . Our results indicate that optimal performance is achieved when a moderate number of objects is used with the support matrix. Otherwise, the results exhibit high variance—while the context matrix provides some improvement, it alone is insufficient to ensure stability.

C. Analyzing the few-shot methods

We compared our new few-shot method (using the Mahalanobis methodology we described before) with the following four established baselines to validate its effectiveness.

- **Single Prototype.** The single-prototype [11] introduces the creation of prototypes to perform classification. This method employs a parametric model, with parameters determined using Maximum Likelihood Estimation (MLE). These parameters, including distances or scores between the sample support set and the computed prototype, are used as the classification threshold for new data points. Additionally, the σ parameter estimated by MLE is important in defining the range for positive observations. The illustration is depicted in Fig. 3a where the Euclidean distance - assuming a normal distribution - creates this circle shape (for a 2D dimensionality) over the data space. The main problem with the Euclidean distance is that it creates a uniform space and can misclassify some samples, as shown in this figure, where q_2 is classified correctly but q_1 is misclassified.
- **Density Prototype.** The density prototype [4] involves leveraging feature embedding across observation dimensions to build density functions. For instance, with an embedding size of m and n observations, m density

TABLE I: The results show the influence of λ , which controls the contributions of Σ_s and Σ_c . Column 1 shows different numbers of images (each potentially containing multiple objects, as indicated in column 2), while the remaining columns display the mean and standard deviation of the results for different values of λ .

# labeled images	# of image crops	$\lambda = 1.00$	$\lambda = 0.75$	$\lambda = 0.50$	$\lambda = 0.25$	$\lambda = 0.00$
1	40 to 54	0.176 (σ : 0.167)	0.195 (σ : 0.159)	0.200 (σ : 0.151)	0.201 (σ : 0.147)	0.202 (σ : 0.144)
2	77 to 105	0.248 (σ : 0.160)	0.251 (σ : 0.139)	0.252 (σ : 0.127)	0.249 (σ : 0.120)	0.245 (σ : 0.115)
3	146 to 172	0.300 (σ : 0.136)	0.285 (σ : 0.108)	0.273 (σ : 0.103)	0.268 (σ : 0.083)	0.265 (σ : 0.068)
4	182 to 218	0.308 (σ : 0.120)	0.289 (σ : 0.093)	0.284 (σ : 0.063)	0.277 (σ : 0.050)	0.270 (σ : 0.041)
5	207 to 288	0.322 (σ : 0.092)	0.306 (σ : 0.051)	0.286 (σ : 0.065)	0.282 (σ : 0.032)	0.274 (σ : 0.029)

TABLE II: This table describes the experimental setup. We compared our method against four SOTA approaches (column 1). Column 2 shows the different object proposal generators we tested. We used four backbone models (one convolutional and three transformers) to generate embeddings, selected based on their performance in the ImageNet benchmark. The final column presents the number of labeled images used in our tests.

Methods	Object proposals	Feature extractor					# labeled images
		Model name	Type	ImageNet Benchmark	# params	Vector length	
Euclidean Prototype	SAM-H	Eff.Net-L2	convolutional	90.2%	480M	5504	1
Density Prototype	HQ-SAM-H	ViT-CLIP	transformers	88.5%	86.6M	768	2
BD-CSPN	FastSAM	Swin V2	transformers	87.1%	88M	768	.
SAM Alone	MobileSAM	XCiT	transformers	85.6%	3M	128	.
Mahalanobis (Ours)	EdgeSAM						
	SlimSAM-50						10

functions are estimated across the n observations, constituting the density prototype. This paradigm mirrors the parametric model of the single prototype method. The proposed embedding is fed into the density functions during inference to derive individual probabilities for each dimension, subsequently aggregated. The model parameters defining the threshold are estimated under the assumption of following a normal distribution to estimate the μ and σ . An upper bound threshold is then utilized to filter the results. Fig. 3b shows how well the density function fits the data; however, if the dataset is small, it can suffer from overfitting and then misclassifying new samples. Fig. 3b shows an example where q_1 and q_2 are misclassified due to overfitting as a result of the density functions.

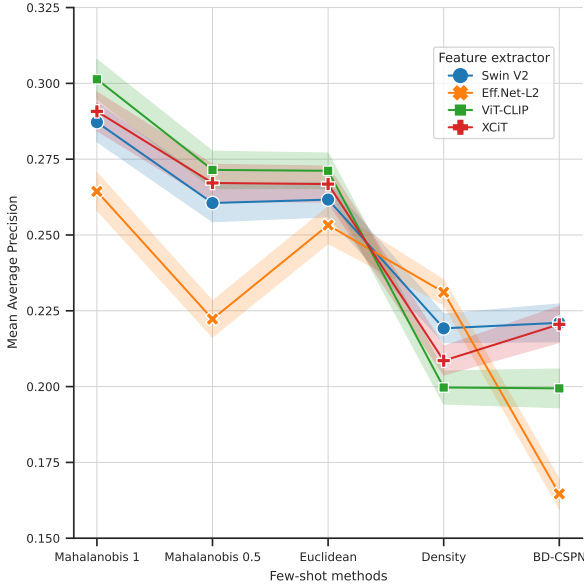
- **BD-CSPN.** The BD-CSPN [35] method is an adaptation of the prototype method for prototype rectification. It incorporates label propagation to mitigate intra-class bias and feature shifting to alleviate cross-class bias. This study focused on a binary classification scenario, where foreground observations constituted one class and background observations formed the other. The cluster’s shape is similar to Fig. 3a, however, a rectification of the centroid has been performed to avoid a biased version of the cluster.
- **SAM Alone.** This method employs SAM-H variation, such as HQ-SAM-H, MobileSAM, EdgeSAM, SlimSAM-50, or FastSAM, to generate object proposals for all objects within the input image. It operates under the assumption that all generated proposals represent

objects, and in our context, since we have only one class in our dataset, this assumption is valid.

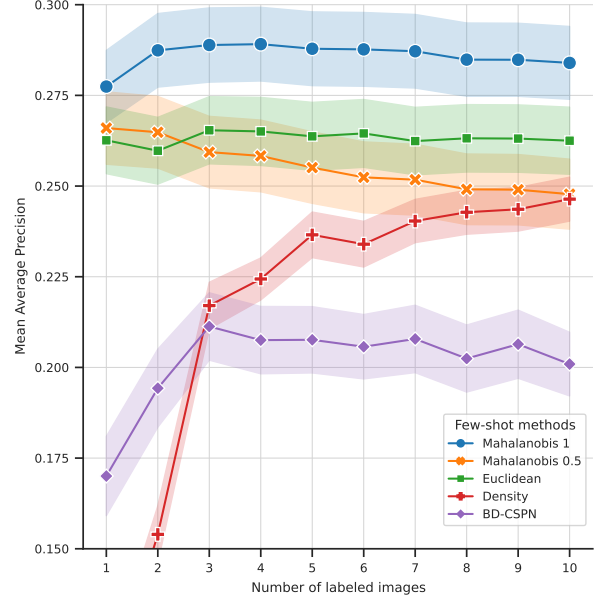
Table II details our experimental design, which evaluated five few-shot methods using six object proposal models and four feature extractors. This setup involved 6000 runs (5 replicates $\times$ 5 methods $\times$ 6 models $\times$ 4 extractors $\times$ 1-10 labeled images). The SAM Alone method was tested in 30 runs (5 replicates $\times$ 6 models). We used five transformer-based and one convolutional-based proposal model, along with three transformer-based and one convolutional-based feature extractor. The final column lists the number of labeled images used for the support set.

Fig. 6 shows the interaction between feature extractors, few-shot methods, and the number of labeled images, highlighting the mean and the variation of the results. Fig. 6a reveals that ViT-CLIP outperforms EfficientNet-L2 for Mahalanobis $\lambda = 0.5$, Mahalanobis $\lambda = 1$, and euclidean prototype methods. However, for the density prototype method, EfficientNet-L2 performs better than transformer-based extractors. With Mahalanobis’ methods, the ranking of feature extractors remains consistent: ViT-CLIP leads, followed by XCiT, Swin V2, and EfficientNet-L2. Mahalanobis $\lambda = 0.5$ with EfficientNet-L2 shows a significant performance drop compared to $\lambda = 1$. The BD-CSPN method generally aligns with density performance across extractors but exhibits a significant performance drop with EfficientNet-L2.

Fig. 6b shows how the density performance relies on the number of labeled images to achieve better performance. This is despite the Mahalanobis with $\lambda = 0.5$, which exhibits a slight decline in performance as the number of labeled



(a) Interaction between few-shot methods and feature extractors.



(b) Interaction between few-shots and the number of labeled images.

Fig. 6: Results of comparing feature extractors and few-shot methods to the left, and the number of labeled images and few-shot methods to the right. We show the interaction plots of these factors and highlight the standard error (shaded region).

images increases, which can be attributed to the inclusion of background information in the context matrix. Meanwhile, the Euclidean prototype and Mahalanobis $\lambda = 1$ few-shot methods maintain consistent performance across varying quantities of labeled data, suggesting that only a few labeled samples are necessary to attain good performance. Notably, the density prototype shows a drastic performance improvement when the number of labeled images increases from two to three. Our Mahalanobis distance outperforms the rest of the implementations, converging fast and with stable results overall.

D. Mahalanobis (ours) against baselines

Given the previous outcomes, for the final experiments we chose a dimensionality reduction to 16 components and $\beta = 1$. Table III showcases the optimal factor combinations identified by maximizing μ and minimizing σ . In this table, we compared our few-shot method with different numbers of labeled images (every image has approximately 40 objects, see Table I - # of image crops), the top feature extractors, and the different object proposal models. To make it short, we show the top 6 object proposal models with the highest feature extractor results.

SAM is used as our baseline model, as it does not perform few-shot classification. The density-based prototype improves performance but requires a large number of labeled samples to avoid overfitting, limiting its effectiveness with a small number of objects. BD-CSPN achieves promising results with as few as two labeled images, which is noteworthy; however, its performance does not surpass that of the simple Euclidean approach. This suggests that adjusting centroids may not be beneficial when only a small number of samples are available.

Interestingly, the standard Euclidean prototype consistently outperforms other baseline methods (excluding ours) when a reasonable number of labeled samples are provided. This highlights that, despite its simplicity, the Euclidean approach remains a strong baseline.

Notably, our Mahalanobis-based method with regularization parameter $\lambda = 0.5$ already outperforms both the Euclidean and other SOTA methods using only a few labeled examples. It is, in fact, the method that achieves the best performance with the least number of samples. Furthermore, with $\lambda = 1$, although it requires slightly more labeled images on average (around three, which is approximately 120 objects), it surpasses all baseline methods, achieving superior results while still maintaining a low annotation burden.

Another key finding is that FastSAM achieves consistently higher precision in detecting pineapples in crops compared to SAM-H, HQ-SAM-H, MobileSAM, SlimSAM-50, and EdgeSAM. Furthermore, EdgeSAM shows lower precision, indicating its limitations in generating accurate object proposals and a higher rate of uncertain detections in pineapple crop environments. The main reason for the good performance of FastSAM is that it was trained using convolutional layers, and the objects we were detecting were not very small (the pineapples have a medium size), consequently, smaller objects that can be detected with the rest of the models were not important for our problem. In other words, FastSAM can detect coarser (medium-sized) objects more effectively than other methods, enabling preliminary filtering of fine-grained (small) objects.

TABLE III: Key insights of the mAP for each method and proposal model. The highest average mAP for each few-shot method is underlined, while the highest average mAP across all few-shot methods is indicated with both underlined and bold. The highest average mAP for each object proposal model among all few-shot methods is highlighted in bold.

Few-shot methods	Num. of labeled images	Feature extractor	Object proposal models	Avg. mAP	σ
SAM Alone	-	-	SAM-H	0.232	0.007
	-	-	HQ-SAM-H	0.219	0.007
	-	-	EdgeSAM	0.077	0.006
	-	-	MobileSAM	0.205	0.007
	-	-	SlimSAM-50	0.145	0.007
	-	-	FastSAM	<u>0.378</u>	0.006
Euclidean prototype	6	EfficientNet-L2	SAM-H	0.335	0.019
	1	ViT-CLIP	HQ-SAM-H	0.287	0.019
	1	ViT-CLIP	EdgeSAM	0.112	0.009
	5	ViT-CLIP	MobileSAM	0.266	0.005
	6	EfficientNet-L2	SlimSAM-50	0.194	0.016
	4	Xcit Nano	FastSAM	<u>0.393</u>	0.002
Density prototype	9	EfficientNet-L2	SAM-H	0.317	0.014
	10	Xcit Nano	HQ-SAM-H	0.270	0.019
	10	Xcit Nano	EdgeSAM	0.102	0.011
	10	Xcit Nano	MobileSAM	0.230	0.012
	9	EfficientNet-L2	SlimSAM-50	0.185	0.013
	8	ViT-CLIP	FastSAM	<u>0.354</u>	0.049
BD- CSPN	2	Swin V2	SAM-H	0.264	0.014
	2	Swin V2	HQ-SAM-H	0.234	0.010
	2	Swin V2	EdgeSAM	0.080	0.007
	2	Swin V2	MobileSAM	0.222	0.010
	2	Swin V2	SlimSAM-50	0.163	0.010
	2	Swin V2	FastSAM	<u>0.379</u>	0.005
Mahalanobis 0.5 (Ours)	3	ViT-CLIP	SAM-H	0.354	0.031
	1	ViT-CLIP	HQ-SAM-H	0.305	0.044
	1	ViT-CLIP	EdgeSAM	0.109	0.011
	3	ViT-CLIP	MobileSAM	0.270	0.010
	1	Swin V2	SlimSAM-50	0.210	0.025
	4	Xcit Nano	FastSAM	<u>0.402</u>	0.004
Mahalanobis 1.0 (Ours)	3	ViT-CLIP	SAM-H	0.389	0.015
	3	ViT-CLIP	HQ-SAM-H	0.346	0.013
	3	ViT-CLIP	EdgeSAM	0.122	0.008
	3	ViT-CLIP	MobileSAM	0.284	0.009
	3	ViT-CLIP	SlimSAM-50	0.243	0.010
	8	Xcit Nano	FastSAM	<u>0.406</u>	0.003

V. CONCLUSIONS

In this paper, we propose a novel pipeline that leverages VFMs and few-shot learning models for OD in drone imagery of pineapple fields, effectively eliminating the need for extensive training. By integrating the proposed Mahalanobis prototype with SVD for scenarios with limited samples, we enhance the performance of SOTA few-shot models. Our results show that increasing the size of the foreground set in the support set improves performance. However, in a small number of label settings, incorporating a background set into the support set yields more stable Mahalanobis prototype estimations. Additionally, dimensionality reduction is critical to counter the sparsity of high-dimensional embeddings when few observations are available, enabling more reliable covariance matrix estimation.

Furthermore, FastSAM—used as the object proposal model—demonstrated superior precision for small objects in pineapple crops, attributed to its YOLO-based architecture. It outperformed the SAM transformer models, particularly the

ViT-masked autoencoder. In contrast, lightweight transformer-based SAM variants showed the weakest performance in this setting. Overall, the proposed methodology significantly reduces reliance on extensive training, making it highly suitable for OD tasks in precision agriculture and other resource-constrained applications.

ACKNOWLEDGMENT

We gratefully acknowledge the National Center for High Technology of Costa Rica (CeNAT), particularly the Colaboratorio Nacional de Computación Avanzada (CNCA), for providing access to the Kabré Cluster. We also thank the University of Costa Rica (Project C4612) for funding, the Master’s program at the Costa Rica Institute of Technology (ITCR) for academic support, and Upala Agrícola for their collaboration and assistance during data collection.

REFERENCES

- [1] Z.-Q. Zhao, P. Zheng, S.-T. Xu, and X. Wu, “Object Detection With Deep Learning: A Review,” *IEEE Transactions on Neural Networks and*

- Learning Systems*, vol. 30, no. 11, pp. 3212–3232, Nov. 2019, conference Name: IEEE Transactions on Neural Networks and Learning Systems.
- [2] Y. Zhang and L. Wu, “Classification of Fruits Using Computer Vision and a Multiclass Support Vector Machine,” *Sensors*, vol. 12, no. 9, pp. 12489–12505, Sep. 2012, number: 9 Publisher: Molecular Diversity Preservation International. [Online]. Available: <https://www.mdpi.com/1424-8220/12/9/12489>
 - [3] J. Dyson, A. Mancini, E. Frontoni, and P. Zingaretti, “Deep Learning for Soil and Crop Segmentation from Remotely Sensed Data,” *Remote Sensing*, vol. 11, no. 16, p. 1859, Jan. 2019, number: 16 Publisher: Multidisciplinary Digital Publishing Institute. [Online]. Available: <https://www.mdpi.com/2072-4292/11/16/1859>
 - [4] F. Fallas-Moya, A. Sadovnik, S. Calderon-Ramirez, and H. Qi, “Object Detection in Pineapple Fields Drone Imagery Using Few Shot Learning and the Segment Anything Model,” in *2023 22nd IEEE International Conference on Machine Learning and Applications (ICMLA)*, Dec. 2023.
 - [5] R. Girshick, “Fast R-CNN,” in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, Dec. 2015.
 - [6] Y. Wang, Q. Yao, J. T. Kwok, and L. M. Ni, “Generalizing from a Few Examples: A Survey on Few-shot Learning,” *ACM Computing Surveys*, vol. 53, no. 3, pp. 63:1–63:34, Jun. 2020. [Online]. Available: <https://dl.acm.org/doi/10.1145/3386252>
 - [7] H. Maciejewski, T. Walkowiak, and K. Szyk, “Out-of-distribution detection in high-dimensional data using mahalanobis distance-critical analysis,” in *International Conference on Computational Science*. Springer, 2022, pp. 262–275.
 - [8] K. Zhou, Z. Liu, Y. Qiao, T. Xiang, and C. C. Loy, “Domain Generalization: A Survey,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 4, pp. 4396–4415, Apr. 2023, conference Name: IEEE Transactions on Pattern Analysis and Machine Intelligence. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/9847099>
 - [9] J. Wang, C. Lan, C. Liu, Y. Ouyang, T. Qin, W. Lu, Y. Chen, W. Zeng, and P. S. Yu, “Generalizing to Unseen Domains: A Survey on Domain Generalization,” *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 8, pp. 8052–8072, Aug. 2023, conference Name: IEEE Transactions on Knowledge and Data Engineering. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/9782500>
 - [10] M. Wang and W. Deng, “Deep visual domain adaptation: A survey,” *Neurocomputing*, vol. 312, Oct. 2018, publisher: Elsevier. [Online]. Available: <https://sciencedirect.proxyucr.elogim.com/science/article/pii/S0925231218306684>
 - [11] J. Snell, K. Swersky, and R. Zemel, “Prototypical Networks for Few-shot Learning,” in *Advances in Neural Information Processing Systems*, vol. 30. Curran Associates, Inc., 2017. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2017/hash/cb8da6767461f2812ae4290eac7cbc42-Abstract.html
 - [12] Z. Zou, Z. Shi, Y. Guo, and J. Ye, “Object Detection in 20 Years: A Survey,” May 2019, arXiv:1905.05055 [cs]. [Online]. Available: <http://arxiv.org/abs/1905.05055>
 - [13] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, “You Only Look Once: Unified, Real-Time Object Detection,” in *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. Las Vegas, NV, USA: IEEE, Jun. 2016, pp. 779–788. [Online]. Available: <http://ieeexplore.ieee.org/document/7780460/>
 - [14] F. Fallas-Moya, M. Gonzalez-Hernandez, and A. Sadovnik, “Size Does Matter: Overcoming Limitations during Training when using a Feature Pyramid Network,” in *2021 20th IEEE International Conference on Machine Learning and Applications (ICMLA)*, Dec. 2021, pp. 1553–1560. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/9679971>
 - [15] J. E. van Engelen and H. H. Hoos, “A survey on semi-supervised learning,” *Machine Learning*, vol. 109, no. 2, pp. 373–440, Feb. 2020. [Online]. Available: <https://doi.org/10.1007/s10994-019-05855-6>
 - [16] Y. Song, T. Wang, P. Cai, S. K. Mondal, and J. P. Sahoo, “A Comprehensive Survey of Few-shot Learning: Evolution, Applications, Challenges, and Opportunities,” *ACM Computing Surveys*, vol. 55, no. 13s, pp. 271:1–271:40, Jul. 2023. [Online]. Available: <https://dl.acm.org/doi/10.1145/3582688>
 - [17] A. Kamilaris and F. X. Prenafeta-Boldú, “A review of the use of convolutional neural networks in agriculture,” *The Journal of Agricultural Science*, vol. 156, no. 3, pp. 312–322, Apr. 2018. [Online]. Available: <https://www.cambridge.org/core/journals/journal-of-agricultural-science/article/review-of-the-use-of-convolutional-neural-networks-in-agriculture/15B08C5CDD85F21EEEF7387536149>
 - [18] T. van Klompenburg, A. Kassahun, and C. Catal, “Crop yield prediction using machine learning: A systematic literature review,” *Computers and Electronics in Agriculture*, vol. 177, Oct. 2020. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0168169920302301>
 - [19] I. Sa, Z. Ge, F. Dayoub, B. Upcroft, T. Perez, and C. McCool, “DeepFruits: A Fruit Detection System Using Deep Neural Networks,” *Sensors*, vol. 16, no. 8, p. 1222, Aug. 2016, number: 8 Publisher: Multidisciplinary Digital Publishing Institute. [Online]. Available: <https://www.mdpi.com/1424-8220/16/8/1222>
 - [20] G. Koch, R. Zemel, and R. Salakhutdinov, “Siamese neural networks for one-shot image recognition,” *ICML deep learning workshop*, 2015.
 - [21] J. Talur, O. Smirnov, and P. Missault, “Few-Shot Out of Domain Intent Detection with Covariance Corrected Mahalanobis Distance,” in *Association for the Advancement of Artificial Intelligence*. Washington, DC: AAAI, 2023.
 - [22] G. J. McLachlan, “Mahalanobis distance,” *Resonance*, vol. 4, no. 6, pp. 20–26, 1999.
 - [23] P. Bateni, R. Goyal, V. Masrani, F. Wood, and L. Sigal, “Improved Few-Shot Visual Classification,” in *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. Seattle, WA, USA: IEEE, Jun. 2020, pp. 14481–14490. [Online]. Available: <https://ieeexplore.ieee.org/document/9157183/>
 - [24] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo, and others, “Segment anything,” *arXiv preprint arXiv:2304.02643*, 2023.
 - [25] R. Deng, C. Cui, Q. Liu, T. Yao, L. W. Remedios, S. Bao, B. A. Landman, L. E. Wheless, L. A. Coburn, K. T. Wilson, and others, “Segment anything model (sam) for digital pathology: Assess zero-shot segmentation on whole slide imaging,” *arXiv preprint arXiv:2304.04155*, 2023.
 - [26] Q. Wu, Y. Zhang, and M. Elbatel, “Self-prompting large vision models for few-shot medical image segmentation,” in *MICCAI Workshop on Domain Adaptation and Representation Transfer*. Springer, 2023, pp. 156–167.
 - [27] Y. Le Cacheux, H. Le Borgne, and M. Crucianu, “Zero-shot learning with deep neural networks for object recognition,” *Multi-faceted Deep Learning: Models and Data*, pp. 127–150, 2021, publisher: Springer.
 - [28] S. Ren, K. He, R. Girshick, and J. Sun, “Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks,” in *Advances in Neural Information Processing Systems*, vol. 28. Curran Associates, Inc., 2015. [Online]. Available: <https://proceedings.neurips.cc/paper/2015/hash/14bfa6bb14875e45bba028a21ed38046-Abstract.html>
 - [29] J. R. Uijlings, K. E. Van De Sande, T. Gevers, and A. W. Smeulders, “Selective search for object recognition,” *International journal of computer vision*, vol. 104, pp. 154–171, 2013.
 - [30] B. D. Carlson, “Covariance matrix estimation errors and diagonal loading in adaptive arrays,” *IEEE Transactions on Aerospace and Electronic systems*, vol. 24, no. 4, pp. 397–401, 1988.
 - [31] J. Fan, Y. Liao, and H. Liu, “An overview of the estimation of large covariance and precision matrices,” *The Econometrics Journal*, vol. 19, no. 1, pp. C1–C32, 2016.
 - [32] A. Mathiasen, F. Hvilshøj, J. Rødsgaard Jørgensen, A. Nasery, and D. Mottin, “What if Neural Networks had SVDs?” in *Advances in Neural Information Processing Systems*, vol. 33. Curran Associates, Inc., 2020, pp. 18411–18420. [Online]. Available: <https://proceedings.neurips.cc/paper/2020/hash/d61e4bbd6393c9111e6526ea173a7c8b-Abstract.html>
 - [33] C. H. Li and S. C. Park, “An efficient document classification model using an improved back propagation neural network and singular value decomposition,” *Expert Systems with Applications*, vol. 36, no. 2, pp. 3208–3215, 2009.
 - [34] L. D. Arias Campos, J. M. Retana Céspedes, D. Torres Sandí, and L. Peña Rivera, “The new golden fruits. production of pineapple and orange in the north zone, costa rica (1974-2015),” *Revista Rupturas*, vol. 8, no. 1, pp. 93–121, 2018.
 - [35] J. Liu, L. Song, and Y. Qin, “Prototype Rectification for Few-Shot Learning,” in *Computer Vision – ECCV 2020*, ser. Lecture Notes in Computer Science, A. Vedaldi, H. Bischof, T. Brox, and J.-M. Frahm, Eds. Cham: Springer International Publishing, 2020.